Chapter 1

Supplementary Sensitivity Analysis

This appendix provides extended sensitivity analyses for both Judge (RQ1a) and Corrector (RQ1b) prompts, quantifying performance stability across prompt variations and reporting the corresponding assumption checks and effect sizes.

1.1 Judge Prompt Sensitivity

To quantify how prompt engineering choices affect judge performance, we perform a sensitivity analysis measuring the relationship between prompt variation (input) and F1 change (output). For judge prompts (RQ1a), prompt distances are computed as cosine distance from the baseline prompt using MPNet embeddings. We use **repeated-measures ANOVA** with CLD×run as subjects and prompt as the within-subject factor, reporting **partial** η^2 as the effect size (proportion of within-subject variance explained by prompt). Bonferroni correction is applied across the four RQ1a tests ($\alpha_{\text{adj}} = 0.0125$).

1.1.1 Derivation: First-Order Sobol Index for a Categorical Factor

For a single categorical factor $A \in \{1, \ldots, K\}$ with $p_a = \Pr(A = a)$, define $\mu_a := \mathbb{E}[Y \mid A = a]$ and $\mu := \mathbb{E}[Y] = \sum_{a=1}^{K} p_a \mu_a$. The first-order Sobol index for A is:

$$S_A := \frac{\operatorname{Var}(\mathbb{E}[Y \mid A])}{\operatorname{Var}(Y)}. \tag{1.1}$$

Since $\mathbb{E}[Y \mid A]$ takes value μ_a when $A = a$, we have

$$\operatorname{Var}(\mathbb{E}[Y \mid A]) = \sum_{a=1}^{K} p_a (\mu_a - \mu)^2. \tag{1.2}$$

By the law of total variance,

$$\operatorname{Var}(Y) = \mathbb{E}[\operatorname{Var}(Y \mid A)] + \operatorname{Var}(\mathbb{E}[Y \mid A]) = \sum_{a=1}^{K} p_a \operatorname{Var}(Y \mid A = a) + \sum_{a=1}^{K} p_a (\mu_a - \mu)^2. \tag{1.3}$$

Hence,

$$S_A = \frac{\sum_{a=1}^{K} p_a (\mu_a - \mu)^2}{\operatorname{Var}(Y)} = \eta^2, \tag{1.4}$$

i.e., the population one-way ANOVA effect size (correlation ratio) for a categorical factor [?]. In the main experiments, we report η_p^2 (partial eta-squared) because our design is blocked by CLD×run (repeated measures), so the sensitivity measure is computed after controlling for these subject-level differences.